

HOW ONLINE PAYMENTS WORK



UPI · CARDS · PAYMENT GATEWAYS



THE DIGITAL PAYMENT REVOLUTION

INTRODUCTION TO ONLINE PAYMENTS ↗

Online payments refer to paying for goods and services electronically via the internet. This method has become popular worldwide for its convenience and speed. From shopping and bill payments to money transfers, digital payments enable the growth of a cashless economy, transforming how we handle financial transactions daily.



TYPES OF ONLINE PAYMENTS

MAIN METHODS USED FOR DIGITAL TRANSACTIONS TODAY

UPI

Unified Payments Interface for instant mobile transfers in India.

DEBIT & CREDIT CARDS

Plastic cards linked to bank accounts or credit lines for online purchases.

DIGITAL WALLETS

Apps like Paytm and PhonePe store money digitally for quick payments.

BANK TRANSFERS

Direct electronic transfer between bank accounts via NEFT, RTGS, or IMPS.



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HOW IT WORKS



WHAT IS UPI?



INSTANT TRANSFERS

India's real-time payment system enables 24x7 money transfers between bank accounts instantly.



VIRTUAL PAYMENT ID

Use mobile number or UPI ID to send and receive money without sharing sensitive bank details.



EASY MOBILE ACCESS

Apps like Google Pay, PhonePe make payments simple with just a few taps on your smartphone.



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DEBIT & CREDIT CARDS EXPLAINED



CARD PAYMENTS



BANK ACCOUNT LINK

Cards connect directly to your bank account (debit) or credit line (credit), enabling instant electronic transactions.



AUTHORIZATION PROCESS

When you pay, the system verifies card details, authenticates via OTP or PIN, then transfers funds securely.



CREDIT VS DEBIT

Debit cards use your own money instantly. Credit cards let you borrow and pay later with interest if not cleared.

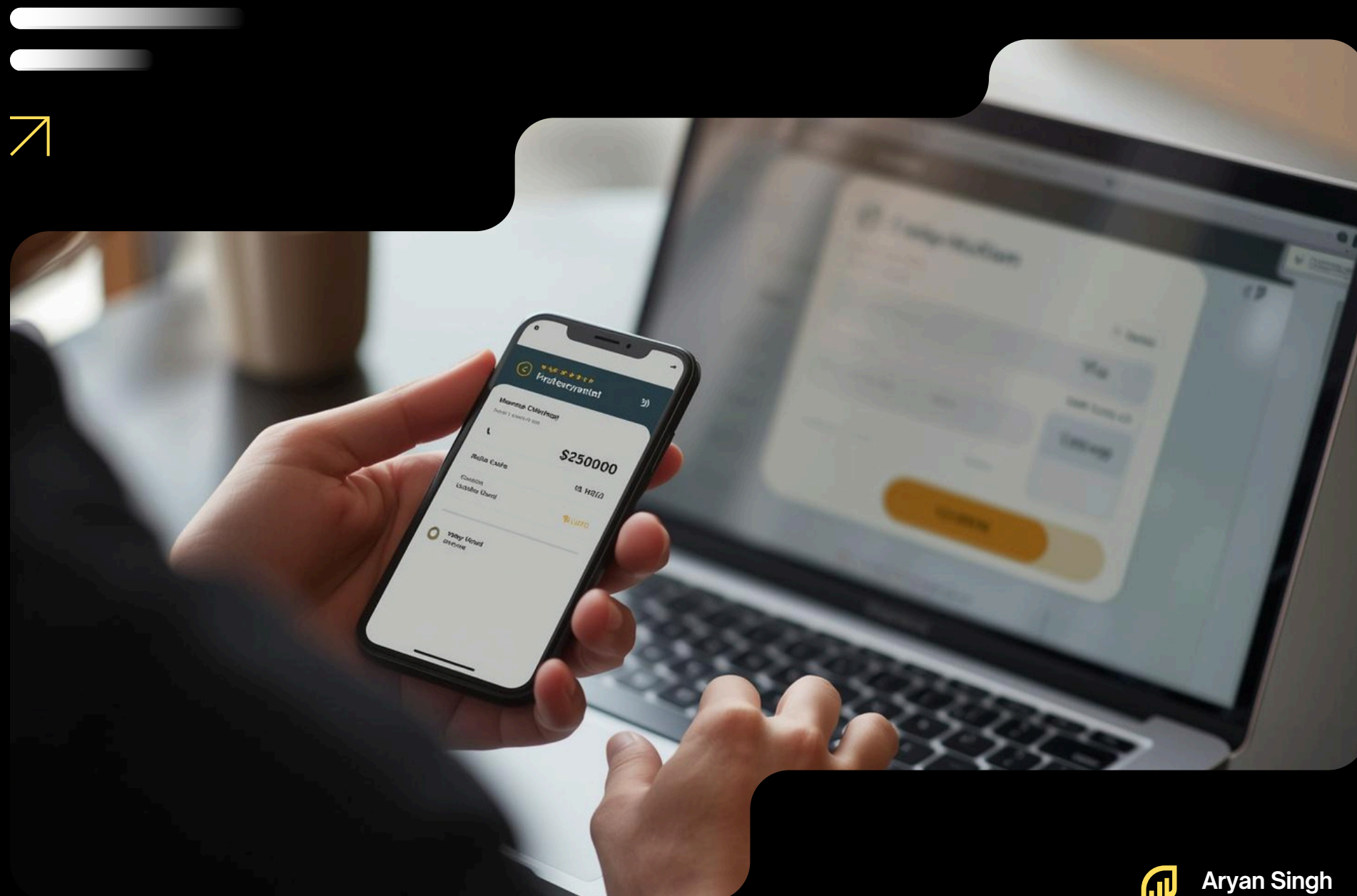
YOUR DIGITAL
CASHIER
ONLINE

WHAT IS A PAYMENT GATEWAY?

A payment gateway is the technology that connects merchants and customer banks. It processes card and UPI payments securely by encrypting sensitive data to protect your information. Acting as a digital cashier, it authorizes transactions, routes payment details, and ensures funds transfer safely between parties.



TRANSACTION FLOW



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STEP-BY-STEP ONLINE PAYMENT PROCESS

CUSTOMER INITIATES PAYMENT

User selects items and clicks pay, entering card details or UPI ID. The payment request is securely sent to the payment gateway for processing.

BANK VERIFICATION & APPROVAL

Payment gateway routes data to the bank server. Bank verifies account balance, authenticates via OTP, and approves or declines the transaction.

FUND TRANSFER & CONFIRMATION

Upon approval, funds are transferred from customer to merchant account. Both parties receive instant confirmation of successful transaction completion.

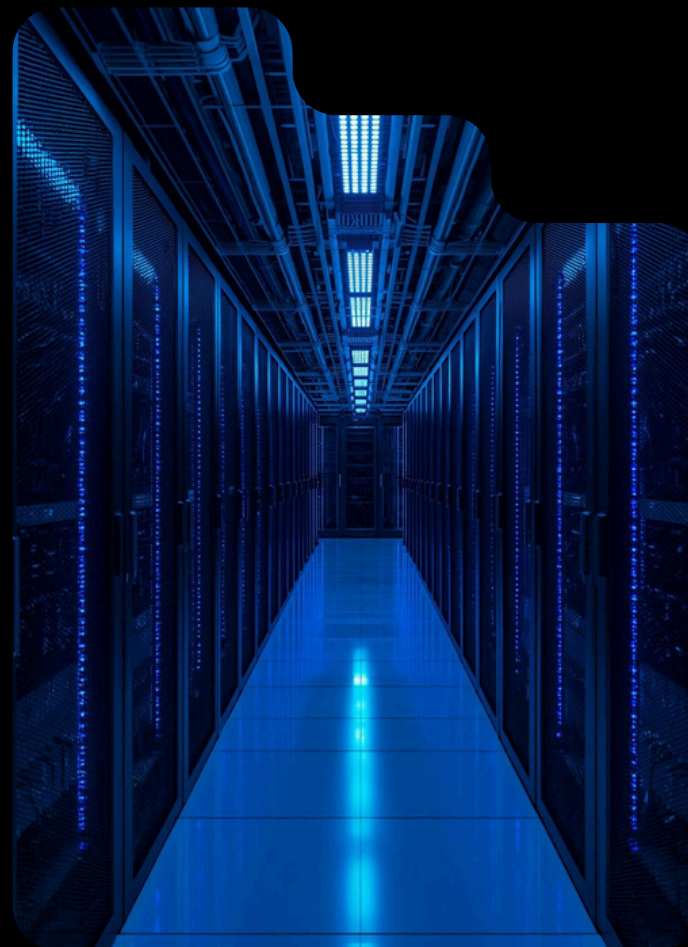
ROLE OF BANKS AND SERVERS



BACKEND INFRASTRUCTURE

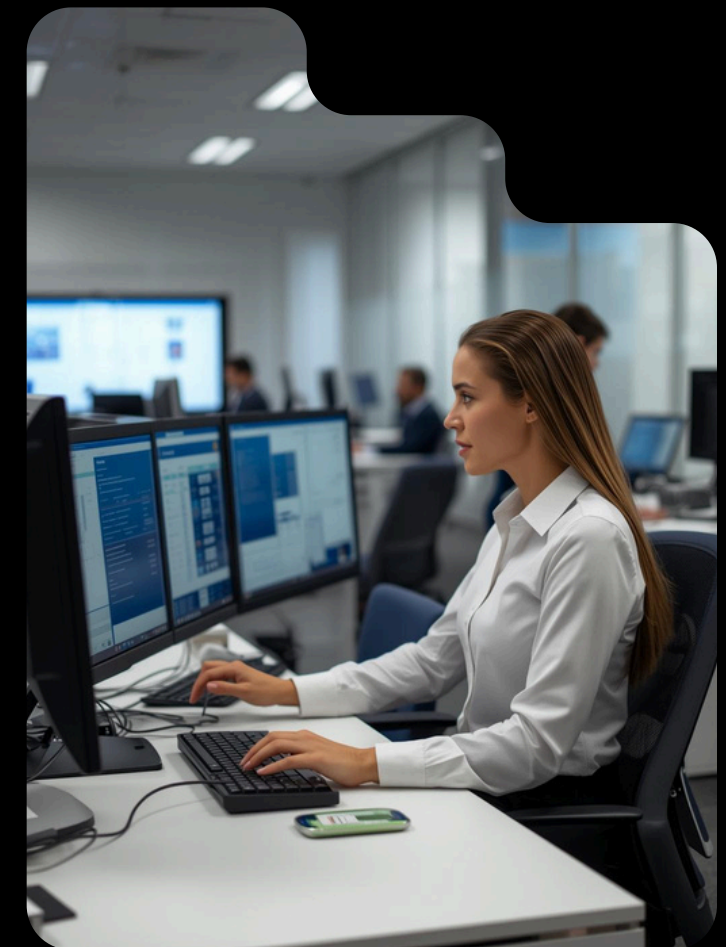
BANK FUNCTIONS

Banks hold customer and merchant accounts, verify balances, authorize transactions, and transfer funds securely.



SERVER OPERATIONS

Servers handle data routing and processing, ensure transaction accuracy and speed, coordinate between payment gateways and accounts.



SECURITY IN ONLINE PAYMENTS



PROTECTING YOUR TRANSACTIONS WITH ADVANCED TECHNOLOGY

OTP AUTHENTICATION

One-Time Passwords sent to your phone verify your identity for each transaction.

DATA ENCRYPTION

Sensitive information is scrambled into unreadable code during transmission.

FRAUD DETECTION

AI-powered systems monitor transactions in real-time to identify suspicious activity.

TOKENIZATION

Card details are replaced with unique codes, keeping your actual data hidden.



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ADVANTAGES OF ONLINE PAYMENTS



KEY BENEFITS OF DIGITAL TRANSACTIONS

CONVENIENCE

Fast transactions completed in seconds from anywhere.

CASHLESS

Eliminates need for physical cash handling and storage.

ACCESSIBILITY

Available 24/7 from any location with internet access.

CONTACTLESS

Tap-to-pay and QR codes enable safe, touch-free payments.



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CHALLENGES AND RISKS



UNDERSTANDING POTENTIAL PROBLEMS IN DIGITAL PAYMENTS

FRAUD & PHISHING

Cybercriminals use fake websites and emails to steal payment credentials.

TRANSACTION FAILURES

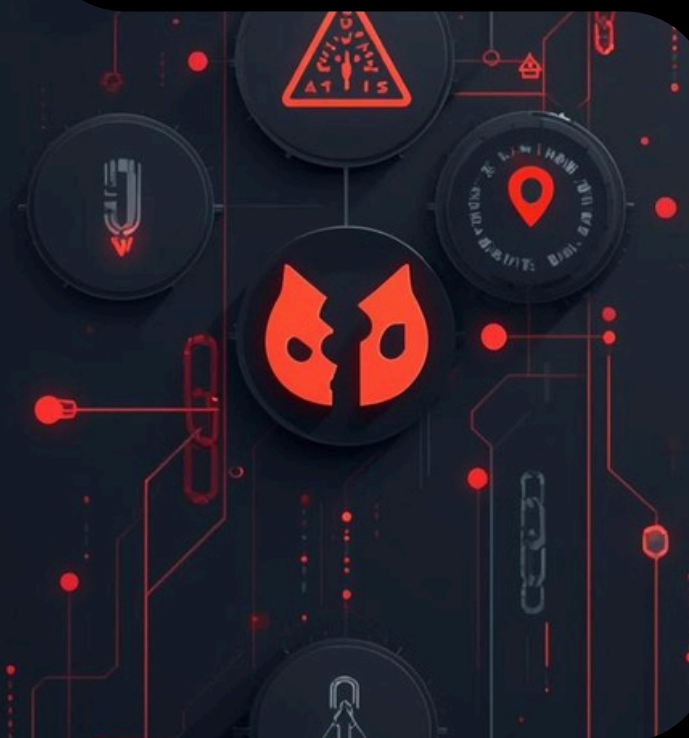
Network issues or server downtime can cause payment delays or failures.

PRIVACY CONCERNS

Sharing personal and financial data raises risks of unauthorized access.

TECH DEPENDENCE

Reliance on internet connectivity and devices limits accessibility.



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POPULAR UPI APPS



REAL-WORLD UPI PLATFORMS

UPI Leaders

Google Pay offers a simple, intuitive interface with a rewards system for transactions. PhonePe boasts wide merchant acceptance across India, making it versatile for everyday payments.

Paytm uniquely combines digital wallet functionality with UPI payments, giving users flexibility. These apps have transformed how millions of Indians handle daily transactions seamlessly.

CARD PAYMENTS



GLOBAL CARD NETWORKS & CONTACTLESS TECHNOLOGY

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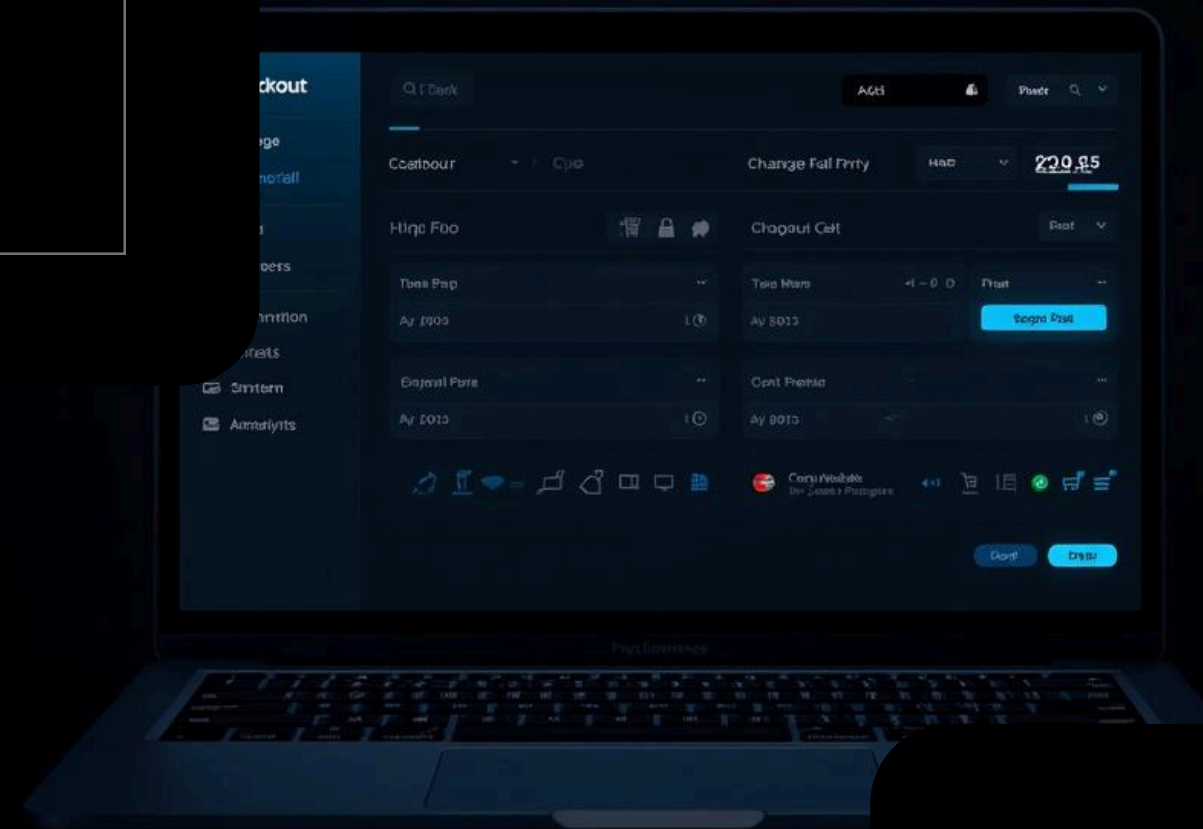
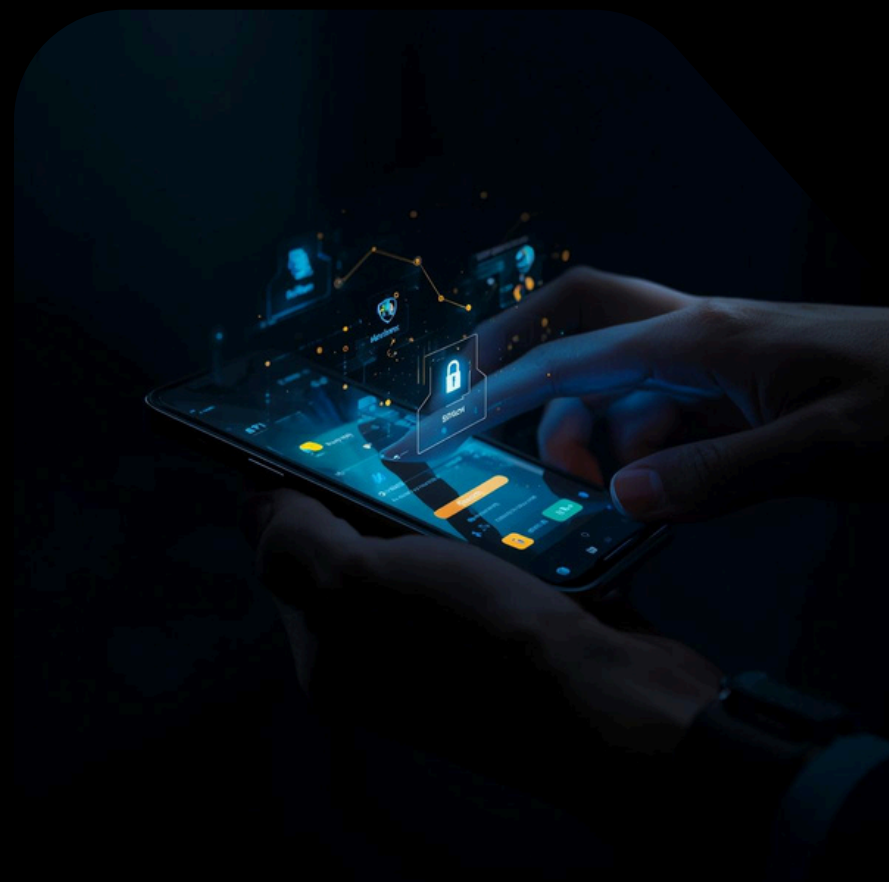
Visa and Mastercard are the world's leading card networks, enabling billions of transactions globally. Their secure infrastructure connects banks, merchants, and customers across 200+ countries.

Contactless NFC cards are revolutionizing payments with tap-to-pay technology. Simply wave your card near the terminal for instant, secure transactions under a spending limit.

PAYMENT GATEWAYS



**TRUSTED
GATEWAY
PROVIDERS
WORLDWIDE**



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In India, Razorpay, PayU, and CCAvenue are leading payment gateway providers. They help businesses of all sizes accept online payments securely and efficiently through multiple payment methods.

Internationally, Stripe and PayPal dominate the market. These platforms enable small businesses and enterprises to accept payments online, supporting global commerce and digital transactions.



UPI VS CARD PAYMENT PROCESS

STEP	UPI FLOW	CARD FLOW
Initiate Payment	✓	✓
Gateway Processing	✓	✓
Bank Verification	✓	✓
Confirmation	✓	✓

TRANSACTION FLOW



Both UPI and card payments follow similar paths through payment gateways to banks. UPI uses VPA for instant transfers while cards require authorization and authentication steps. The gateway securely routes data between customer and merchant banks.

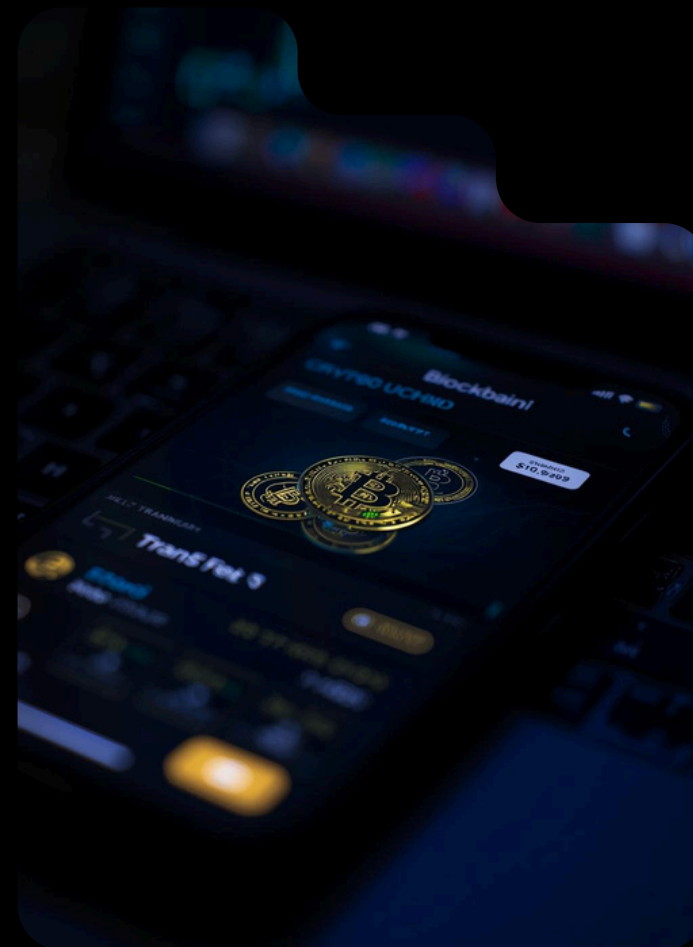
EMERGING TRENDS IN DIGITAL PAYMENTS



FUTURE OF PAYMENTS

AI FRAUD DETECTION

Machine learning algorithms analyze transaction patterns in real-time to detect and prevent fraudulent activities instantly.

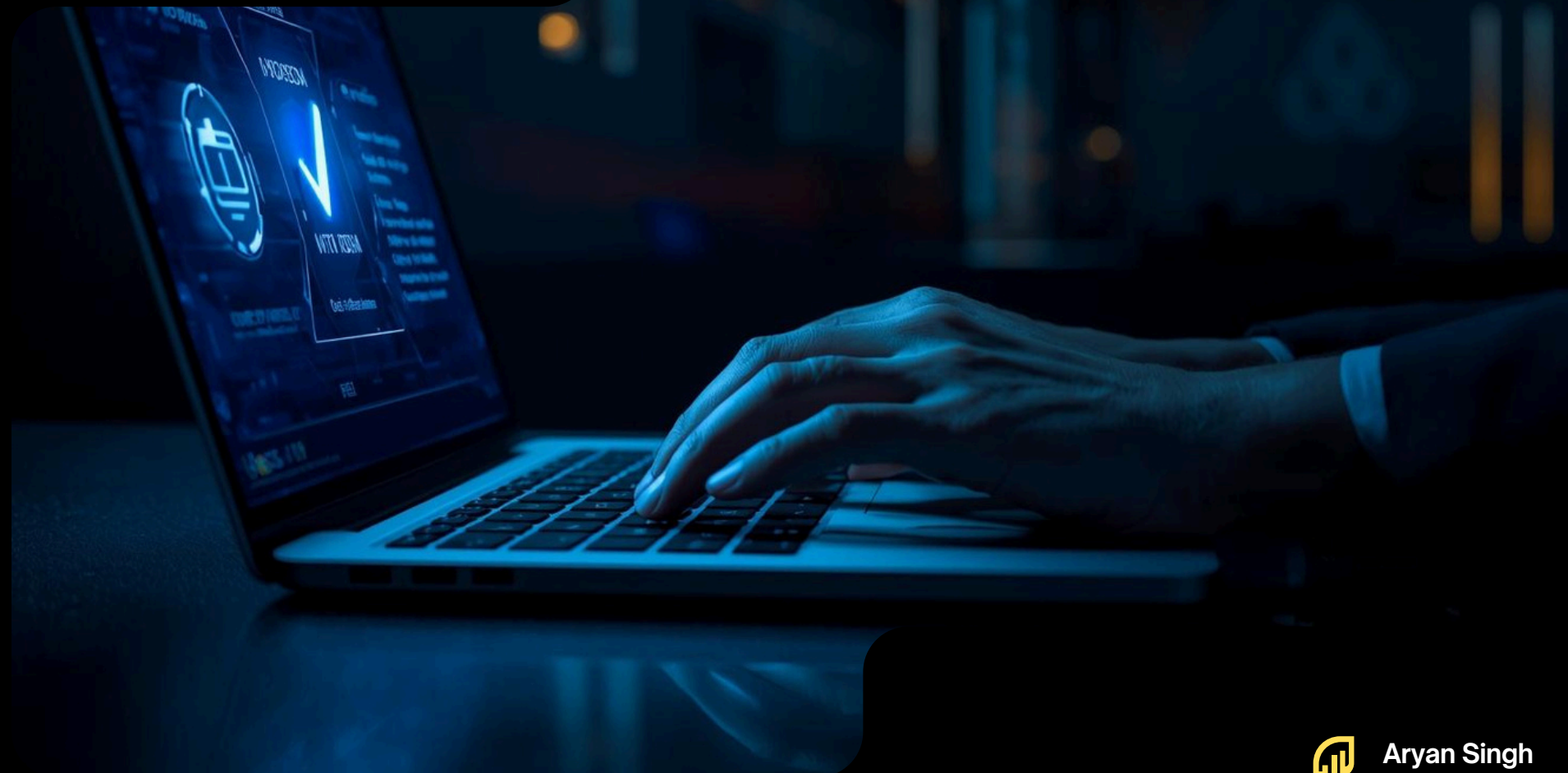


BIOMETRIC PAYMENTS

Fingerprint scanning, facial recognition, and voice authentication are replacing traditional PINs for faster, more secure transactions.



SUMMARY



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KEY TAKEAWAYS FROM ONLINE PAYMENTS

FAST & SECURE TRANSACTIONS

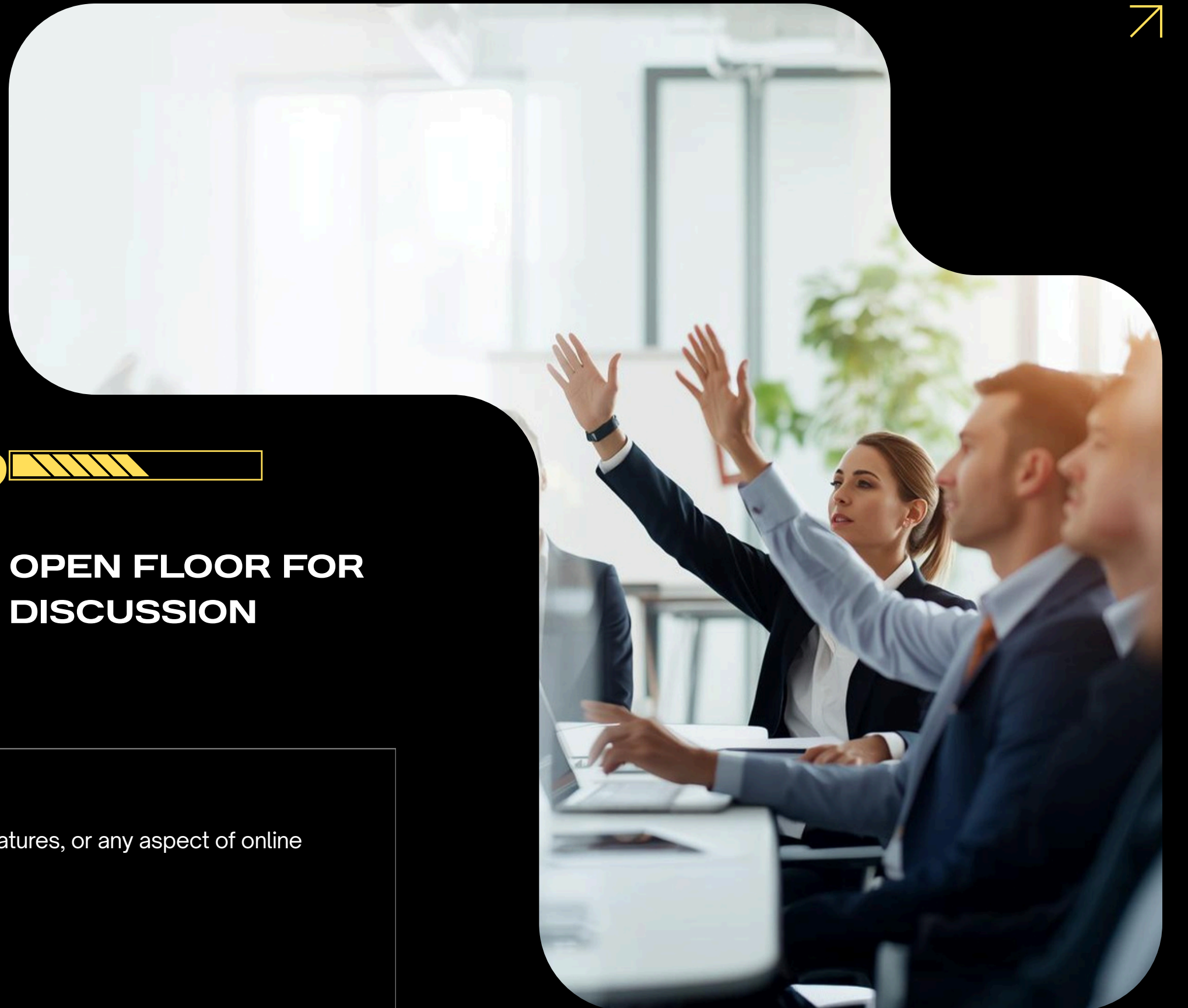
Online payments enable instant, secure cashless transactions that power modern commerce, from shopping to bill payments to money transfers.

MULTIPLE PAYMENT METHODS

Choose from UPI, debit/credit cards, digital wallets, and payment gateways—each offering unique benefits for different transaction needs.

SECURITY IS CRITICAL

OTPs, encryption, tokenization, and fraud detection systems work together to protect your financial data and ensure safe digital transactions.



QUESTIONS?

**OPEN FLOOR FOR
DISCUSSION**

Feel free to ask about UPI, card payments, payment gateways, security features, or any aspect of online transactions we covered today.

ADDITIONAL RESOURCES



CONTINUE YOUR LEARNING JOURNEY

UPI Official Resources:

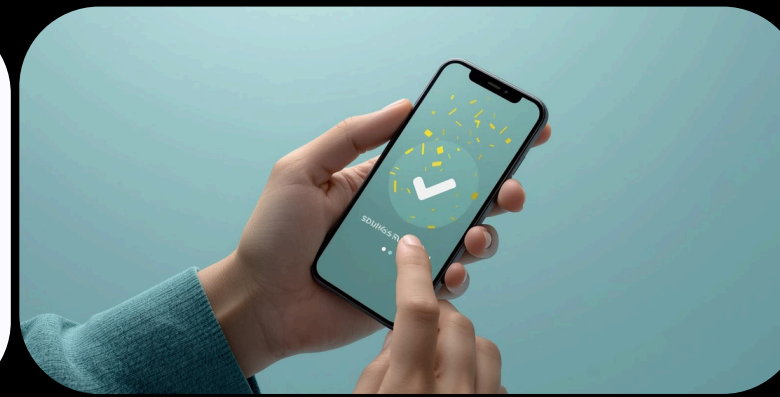
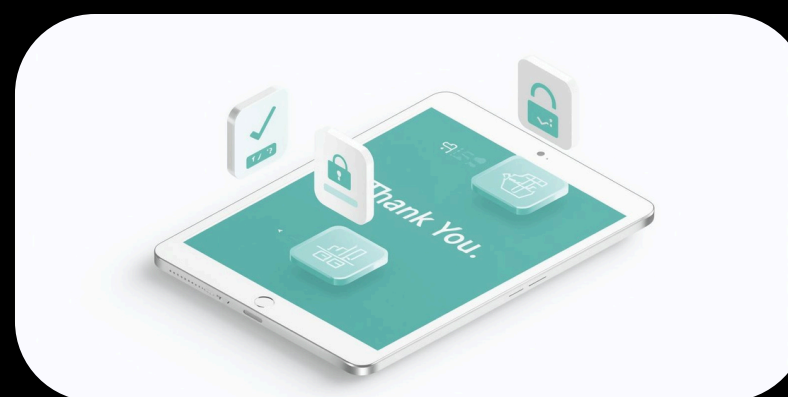
- NPCI UPI Portal: npci.org.in/upi
- BHIM App Guide: bhimupi.org.in
- RBI Digital Payments: rbi.org.in/digitalpayments

Card Security & Gateway FAQs:

- **Visa Security Center:** visa.com/security
- **Mastercard Safety Tips:** mastercard.com/safety
- **Razorpay Docs:** razorpay.com/docs

THANK YOU!

QUESTIONS? LET'S DISCUSS!



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Presented by Aryan Singh Sikarwar. Feel free to reach out for questions or further discussion on online payments.